

Team **Androids**

Business Entity Resolution · 10 million records, 1.7 million businesses

PKT

Pardheev Krishna Tammineni

MT

Monika Tammineni

PH

Pallam Hemachand



The Scale Challenge

Each of almost 10 million records must find its one business, or none, among 1.7 million.



9,969,589 records
incoming, to be linked

which one, or none?

US

India

France

0 labels



1,732,544 businesses
the reference list

6.7 trillion possible same-country pairs

7.10 per business candidates we keep: under 2 in every million pairs

Meet Three Records

Real records are messy, and the metric punishes every wrong link.

🔄 we will meet them again

📄 US

**Legacy Marketing
Enterprises (Inc)**

(no address)

no address: only the name

📄 India

**विजय एंटरप्राइजेज
प्राइवेट लिमिटेड**

24/11, Mumbai, महाराष्ट्र
*(Vijay Enterprises Private
Limited)*

Hindi script, short address

📄 France

TQ COMITE SARL

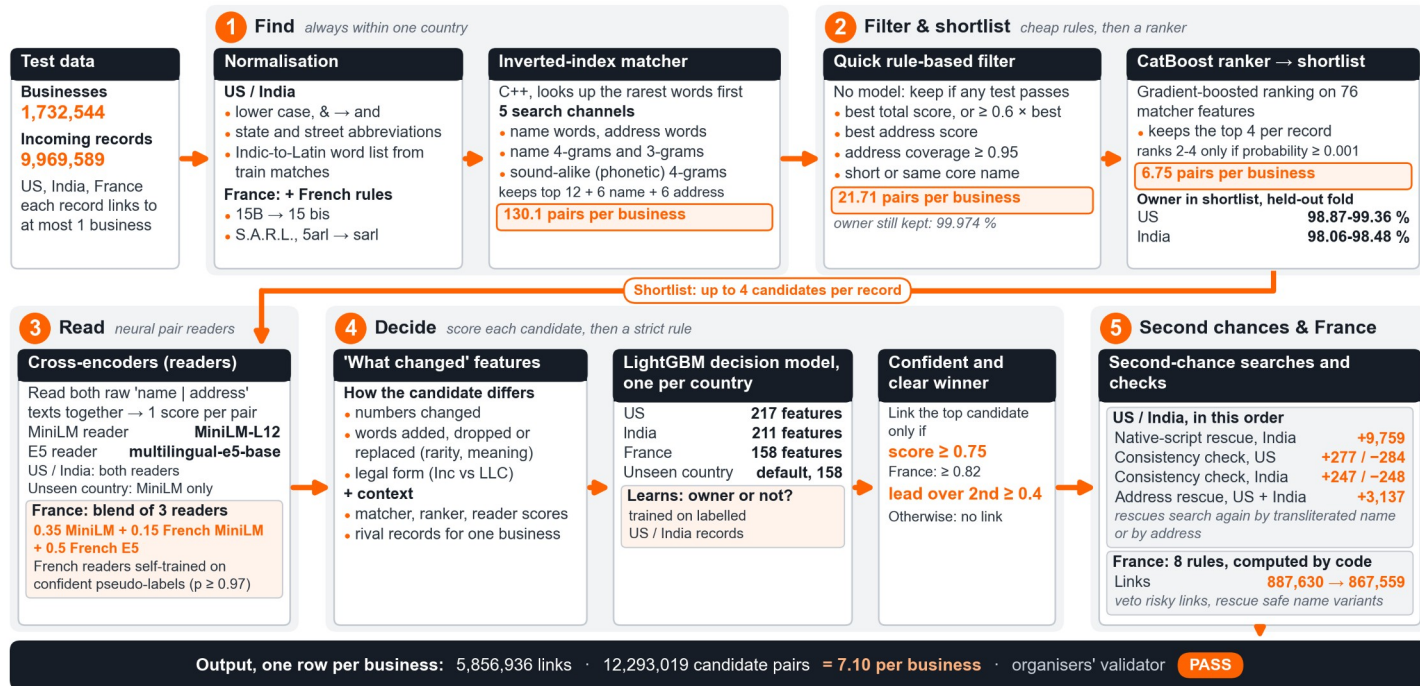
2 R SUZANNE LENGLEN,
NANTES

no labels in France

Score, per business (F0.5): a wrong link costs more than a missed one.

Pipeline Architecture

Five zones per country, from cheap to expensive.



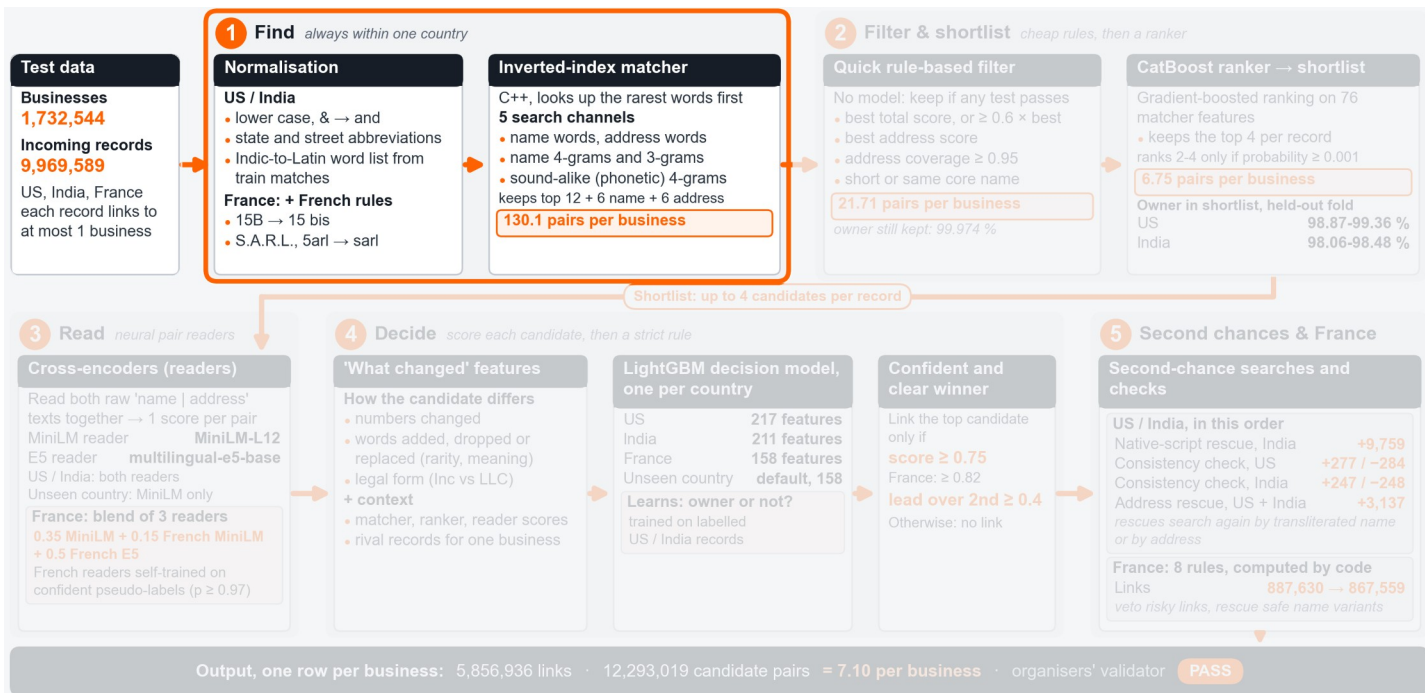
1 Find
2 Filter
3 Read
4 Decide
5 Second chances

Cheap steps first;
 costly models read
 only the shortlist.

Runs separately
 per country.

Architecture: Find

We clean the text, then look up businesses that share rare words, letter pieces or sounds.

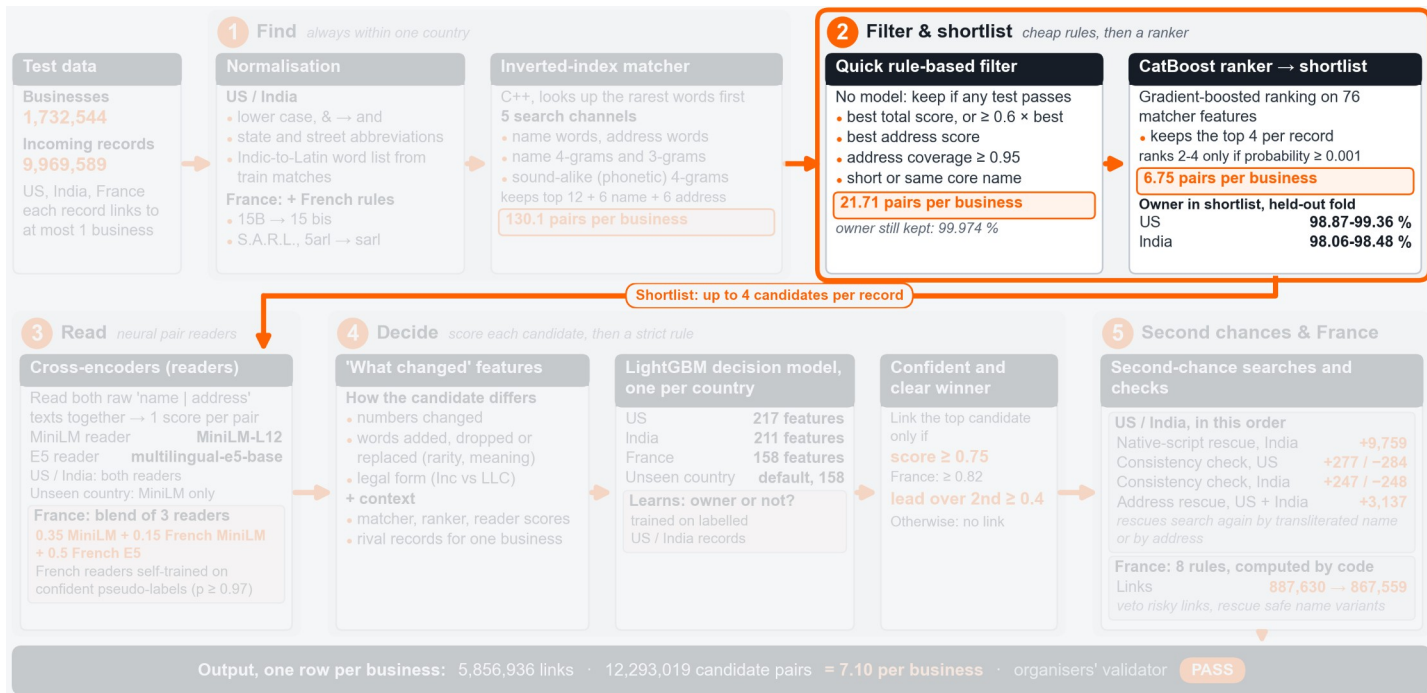


Find

Typos and abbreviations still find their business.

Architecture: Filter

A quick rule and a ranker cut the list to at most 4 candidates per record.



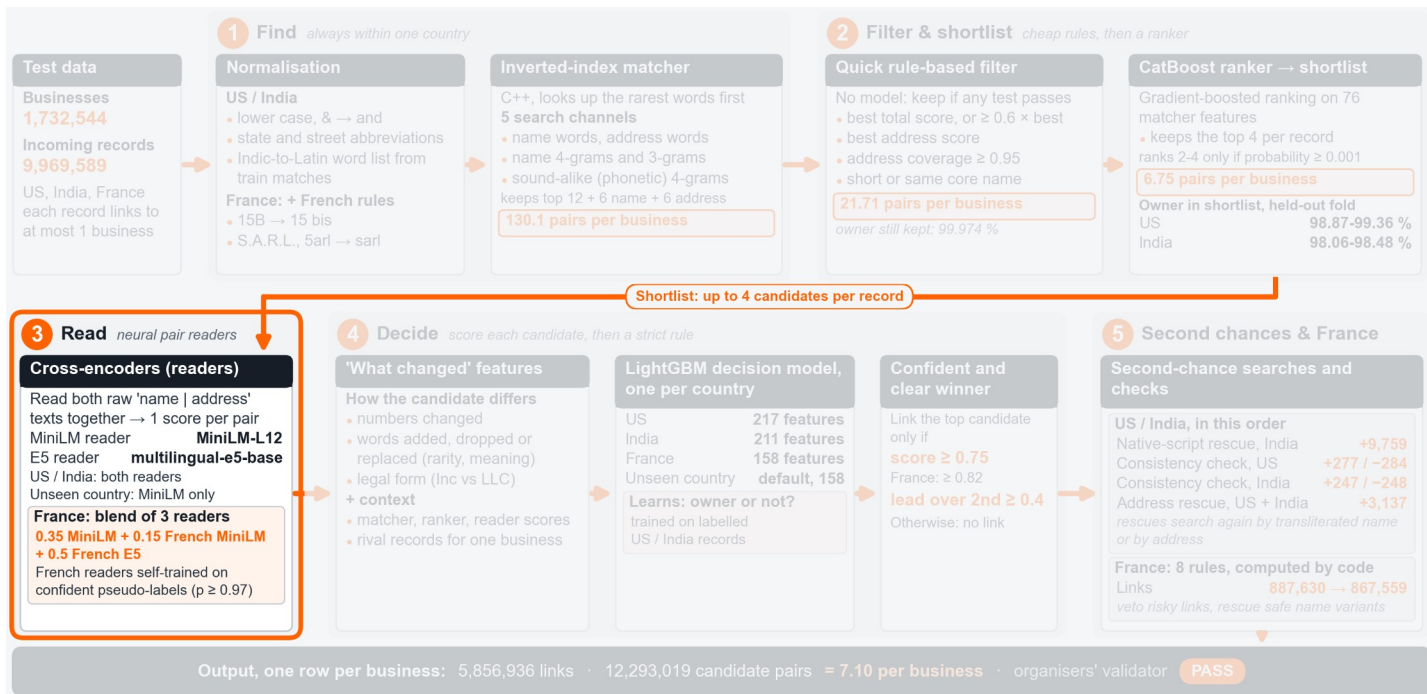
2

Filter & shortlist

The shortlist is all the expensive models ever see.

Architecture: Read

Cross-encoders (we call them readers) read the record and each candidate side by side.



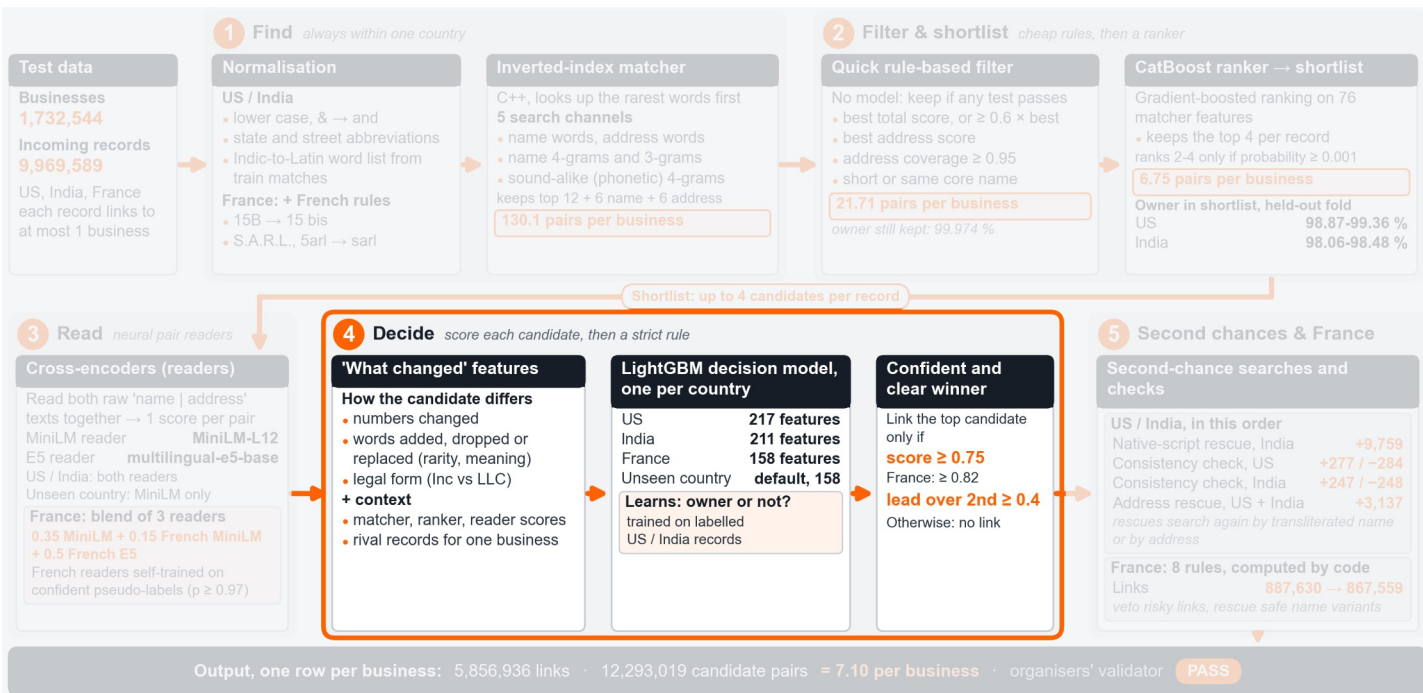
3

Read

Reading both
names together
catches
Inc vs LLC.

Architecture: Decide

We link only a confident, clear winner.



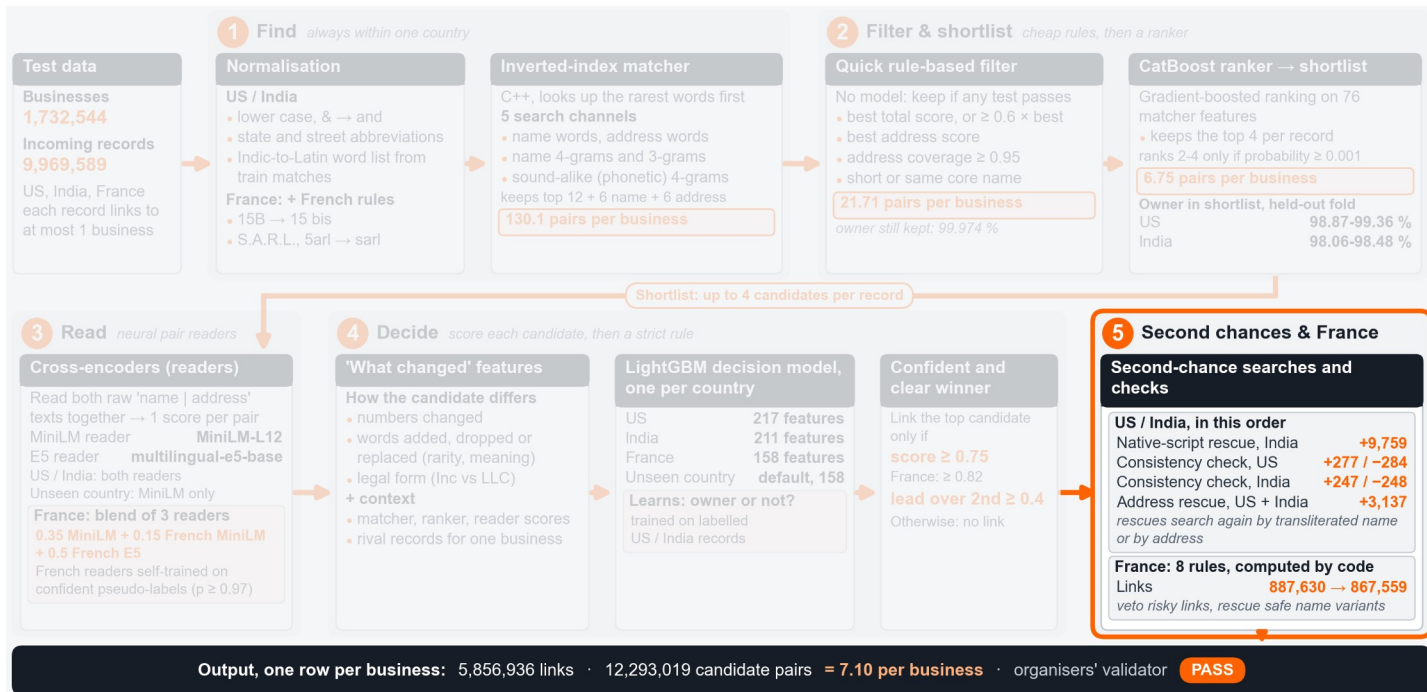
4

Decide

Score ≥ 0.75 and
 ≥ 0.4 ahead of
 the runner-up.

Architecture: Second Chances

Unlinked records get a second search, and France gets its own rules.



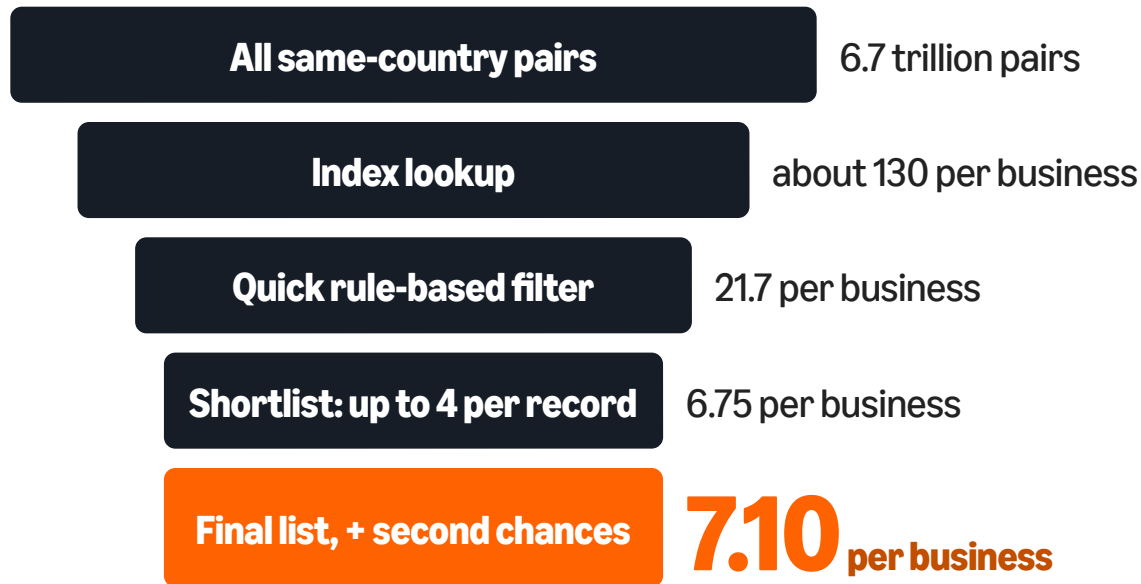
5

**Second chances
& France**

Second
searches add
missed links;
France rules
block risky ones.

Blocking: the Funnel

The expensive models read only about 7 candidates per business, so cost grows in a straight line.



Scaling to billions

US**India****France**

today: one index per country



next: smaller pieces by state or
postcode, run in parallel

What Changed

We describe every candidate by exactly how it differs from the record.

Record: **Legacy Marketing Enterprises (Inc)**

Legacy Marketing Enterprises **LLC**

legal form changed

Legacy Marketing **Industries** Inc

1 word replaced

Also counted: numbers changed, words added or dropped.

Reader scores
+ "what changed"
+ lead over rivals

⚙️ **LightGBM**
decision model,
one per country

Link if confident AND a clear winner

≥ 0.75

score

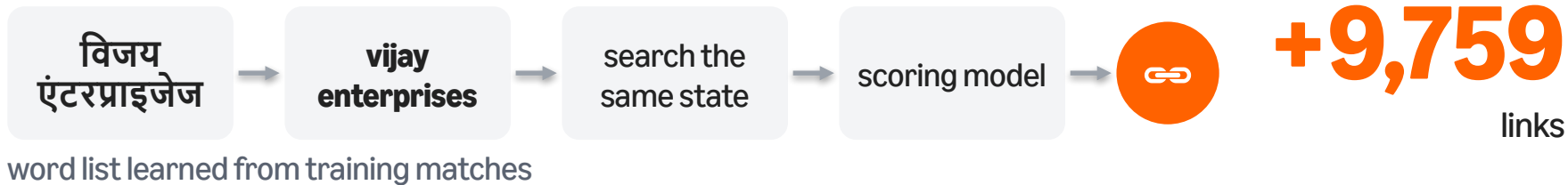
≥ 0.4

lead

Second-Chance Searches

When the first search misses the right business, unlinked records get a second search.

Names in Indian scripts

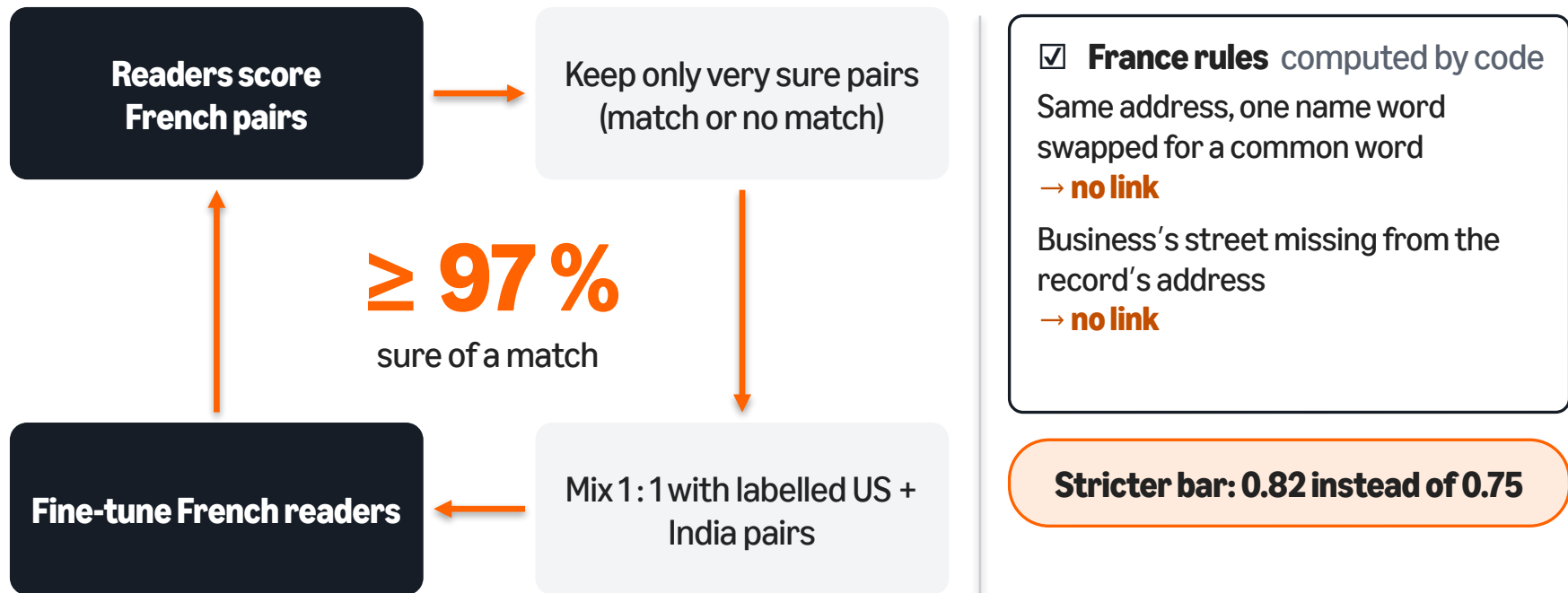


Search again from the address



France: Zero Labels

With no French labels, our readers taught themselves, and rules computed by code add caution.



Training and Selection

We tested changes on businesses the models had never seen.

10

folds

Businesses split into 10 folds (each business's records stay together)



train ranker + readers

train decision models

orange folds: hidden test, in turn

1 Score it on a hidden fold



2 New readers: written checks



✓ **Keep or drop**

Some later choices and the final pick also used public-leaderboard scores.

Handling Edge Cases

Businesses with no records, unseen countries and noisy spellings each get a cautious default.



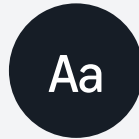
No matching record

Link only on a clear win.



Unseen countries

Default route built like France. Untested.



Noisy spellings

Clean-up, letter pieces, sounds, second chances.

100,087

businesses left with no link, on purpose

Example: the US

No address: the legal form and one word decide.

 US

Legacy Marketing Enterprises (Inc)

(no address)

decision model score, Inc →

0.965

1 Find

Similar names
found; no
address to help

2 Shortlist

Top 4 kept: Inc,
LLC, PC,
Industries

3 What changed

LLC, PC: legal
form; Industries:
one word

4 Read & score

Inc scores
highest; the
others far lower

5 Decide

**Confident and
clear winner →
linked**

 **Linked to Legacy Marketing Enterprises Inc, Windsor, CT**

Example: India

A Devanagari name is found by its second-chance search.

India

विजय एंटरप्राइजेज प्राइवेट लिमिटेड
24/11, Mumbai, महाराष्ट्र

(Vijay Enterprises Private Limited)

second-chance model score →

0.99994

1 Find

Same-name businesses found, but in other places

2 Shortlist & score

All 3 score near zero

3 Decide

Not confident → left unlinked

4 Second chance

Read as 'vijay enterprises', searched in Maharashtra

5 Confirm

Only one with house no. 24/11 → linked

🔗 **Linked to Vijay Enterprises Private Limited, Ghatkopar, Mumbai**

Example: France

A confident link is vetoed, because abstaining is cheaper than a wrong link.

France

TQ COMITE SARL

2 R SUZANNE LENGLEN, NANTES

0.9998 → veto

1 Find & shortlist

2 kept: same
address, or
same name

2 French readers

French readers
reject the same
name in Pessac

3 Decide

Confident and
clear winner: TQ
Parents, Nantes

4 France rule check

'Parents'
became 'Comite'
at the same
address

5 Veto

**Rule vetoes the
link → left
unlinked**

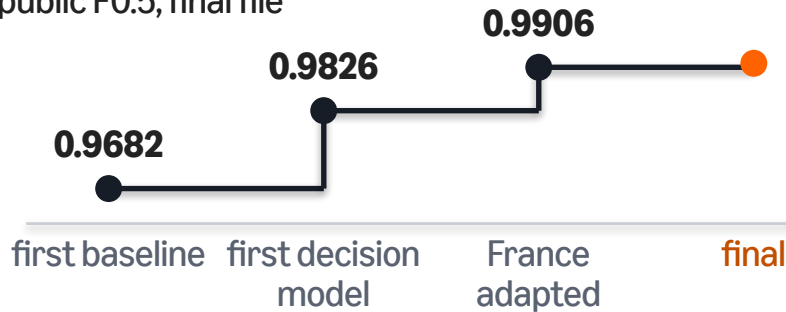
✗ **Left unlinked; both candidates stay in the candidate file**

Results and Learnings

0.9913 on the public leaderboard, with hidden-fold scores in the same range, and honest limits.

0.9913

public F0.5, final file



Hidden-fold F0.5 (fold 9 / fold 8)

US 0.9912 / 0.9915

India 0.9900 / 0.9903

France no labels, so no fold score

- ✓ **Worked:** describing exactly what changed
- **Next:** untouched check fold, small French label set

Limits We never had a fully untouched test of the final file: the hidden folds also helped set our thresholds. France relied on self-training and on rules designed after looking at unlabelled test records, and we kept the French E5 reader although it failed 2 of its 4 written checks. The public leaderboard helped us pick the final file.

Thank you Questions

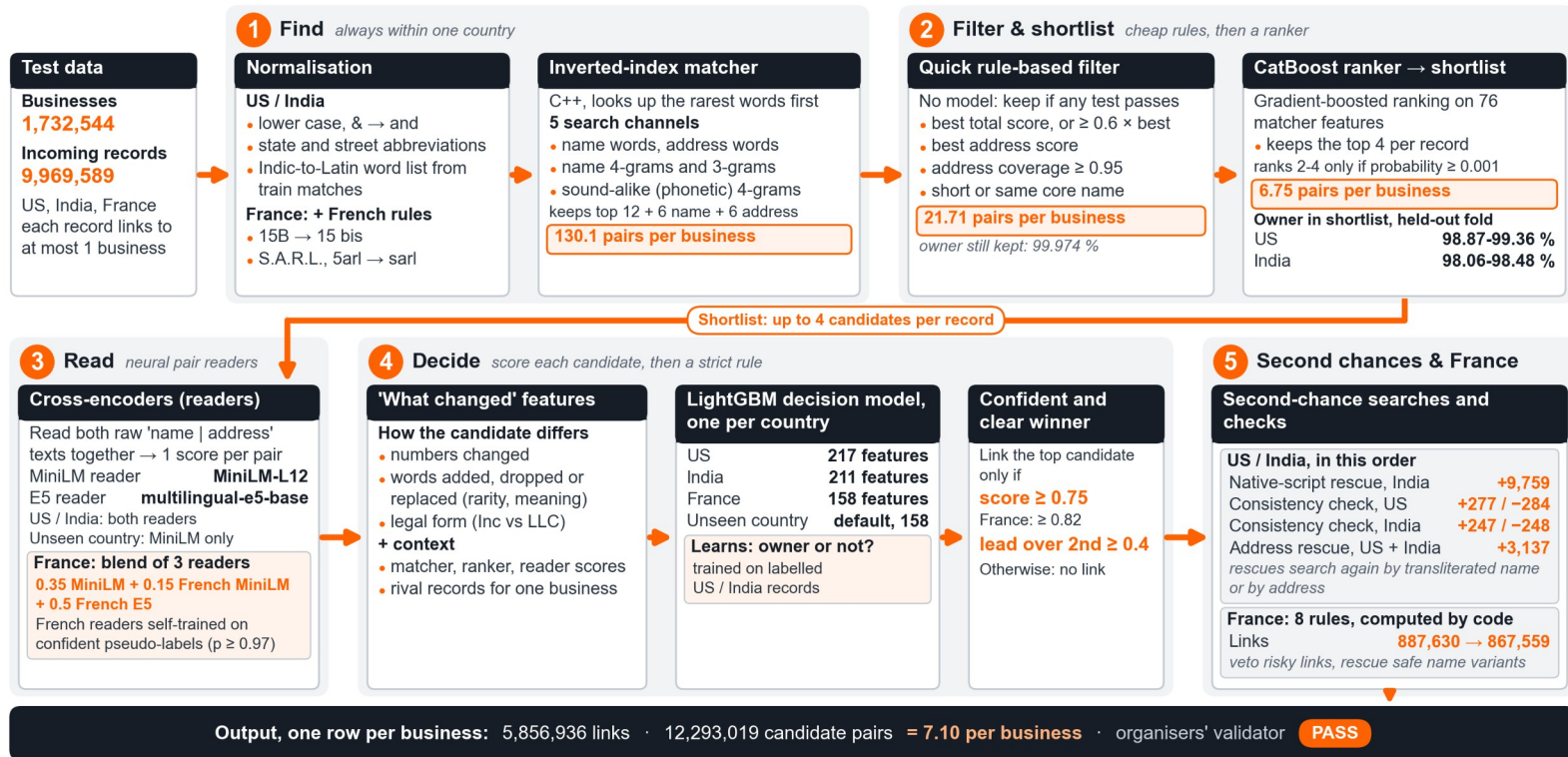
Team Androids

6.7 trillion possible pairs → 12.3 million candidates · public F0.5 0.9913



Full Pipeline Architecture

Backup · the complete diagram



Funnel in Numbers

Backup · every step of the search, per business

Step	Pairs	Per business
All same-country pairs	6,724,569,566,212	-
Index lookup (blocking study, 25 Sep)	-	130.1
Quick rule-based filter	37,614,782	21.71
Shortlist (top 4 per record)	11,687,317	6.75 (derived)
Candidate file (+ second-chance searches)	12,293,019	7.10

Filter cost: true business kept for 99.974 % of held-out records; fold-9 F0.5 0.990117 → 0.990086.

Recall on train fold 9: index 99.37 % US / 98.51 % India; shortlist at least 98.87 % / 98.06 %.

Per record: 1.23 candidates; 1.17 reach the readers (derived).

Held-Out Scores

Backup · F0.5 on hidden folds, without records touching the scored fold

Decision model, as measured	Fold 9	Fold 8
India: final recipe with 3 of its 5 seeds, consistency check	0.990046	0.990298
US: the same 3-seed model, consistency check (the final US model trains on US rows only and adds 6 features)	0.991212	0.991545
France recipe tried on US / India data at the France threshold (a stand-in check, not a French score)	0.990508 / 0.989467	0.990977 / 0.989850

Fold 9 scored by fits on folds 6-8, fold 8 by fits on 6, 7 and 9. Excludes second-chance searches and France rules.

Public score of each adopted upload: 0.968212 → 0.969966 → 0.970125 → 0.982561 → 0.990556 → 0.990621 → 0.990676 → 0.990748 → 0.990788 → 0.990881 → 0.990934 → 0.991114 → **0.991300**

Written Checks

Backup · the French E5 reader failed 2 of 4 checks; kept for the best public score

French E5 reader: check	Result	Outcome
US / India quality within tolerance	AUC 0.998544 vs 0.998570	pass
Agreement with held-out French pseudo-labels	0.999206 (needed 0.98)	pass
Generated labelled French test set	best F0.5 0.910516 vs 0.914964	fail
Expected gain must be safely positive	pessimistic -0.0001806	fail

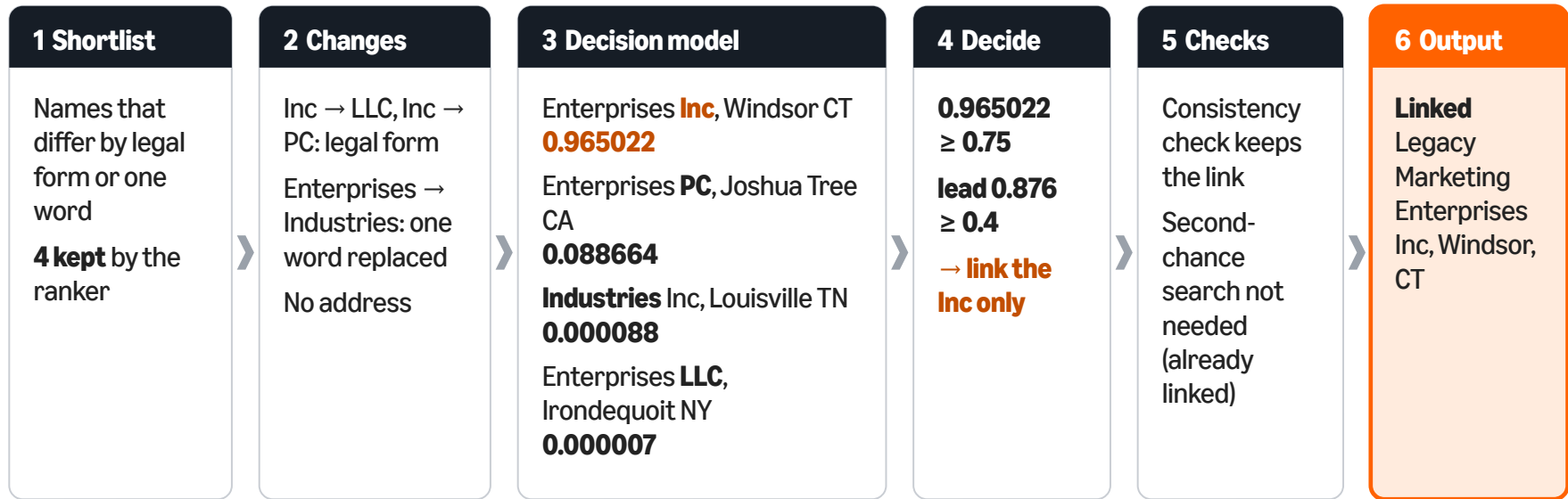
Change	Evidence	Decision
Quick rule-based filter	fold-9 F0.5 0.990117 → 0.990086	kept
US / India decision models refit on folds 6-9	public 0.990621 → 0.990676	kept
France decision model refit	public 0.990726 vs 0.990748	dropped
Second French self-training round	public 0.991277 vs 0.991300	dropped
Larger XLM-R reader	development tests	dropped

US Example in Detail

Backup · No address: legal form and one word separate four candidates

“Legacy Marketing Enterprises (Inc)” · no address

US. Test labels are private: this shows how the pipeline decides, not whether it is right



India Example in Detail

Backup · A Devanagari name the Latin index could not place

विजय एंटरप्राइजेज प्राइवेट लिमिटेड · 24/11, Mumbai, महाराष्ट्र

India. Latin reading: Vijay Enterprises Private Limited · 24/11, Mumbai, Maharashtra

1 Shortlist	2 Model	3 Decide	4 Search key	5 Second-chance model	6 Output
3 × "Vijay Enterprises Private Limited" Agra (UP), Satara, Byculla Ranker top probability 0.0028	Agra 1.9e-6 Satara 6.9e-6 Byculla 7.2e-6	All below 0.75 → unlinked goes to the second-chance search	Word list from training matches: विजय → vijay एंटरप्राइजेज → enterprises Legal words dropped vijay enterprises, Maharashtra	10 businesses with that name in Maharashtra → top 5 Ghatkopar, house no. 24/11: 0.99994 Next (Bhandup West): 0.0000191	0.99994 ≥ 0.88 lead 0.99992 ≥ 0.3 → link added

France Example in Detail

Backup · No labels: French readers, then rules computed by code

“TQ COMITE SARL” · 2 R SUZANNE LENGLEN, NANTES

France. No labels: the French readers are self-trained on confident pseudo-labels

1 Shortlist

Search 16 →
quick filter
keeps 2

TQ Parents
SARL, Nantes
(ranker 0.9991)

TQ Comite
SARL, Pessac
(0.0420)



2 Readers

Score	Nantes	Pessac
MiniLM	7.5000	7.1250
Fr. MiniLM	8.9375	-1.9219
Fr. E5	10.8125	-10.8125
Blend	9.3719	-3.2008

Blend = 0.35 MiniLM + 0.15 French
MiniLM + 0.5 French E5 reader



3 Model

Nantes
0.999837
Pessac
0.000098



4 Decide

Accept TQ
Parents
0.999837
≥ **0.82**
lead
0.999740
≥ **0.4**



5 France rules

Same no. 2 and
street (match
100)
One word
swapped:
“parents” →
“comite”
Second rule
agrees
→ **link vetoed**



6 Output

**Record
unlinked**
Both
businesses
stay in the
candidate file